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RESEARCH ARTICLE

Simplicity is key: restoration protocols for nonregenerating forests degraded by overabundant herbivores

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Global forests are being lost and degraded at an alarming rate; hence ecological restoration becomes an integral component ensuring future forest health. Beneficial effects of restoration will depend on scientifically based practices within an adaptive management framework. On the island of Newfoundland, moose (*Alces alces*) have become overabundant since their introduction in the early 1900s causing regeneration failure of the balsam fir (*Abies balsamea*) forest. Intensive selective browsing by moose on this foundation species within naturally insect-generated gaps has created "spruce-moose meadows." Experimental restoration to support Parks Canada Ecological Integrity targets was implemented in the boreal forest of Terra Nova National Park (Newfoundland, Canada), along a gradient of disturbance from closed canopy forest to large insect-disturbed gaps. Seedling planting was carried out under various ground preparation treatments (field planting, aboveground suppression, and scarification). Seedling performance (survival, growth, and browsing occurrence) was monitored over 2 years and mixed-effects models were constructed to determine seedling responses, which form the template of future forests. Results show minimally positive effects of the ground treatments along the gradient of disturbance, while environmental conditions and seedling individual traits explained the majority of seedling responses. Better growth, lower survival, and higher browsing intensity were observed with increasing forest disturbance, with taller seedlings at planting performing the best. Considering that no substantial biological benefits were detected following ground treatments, which are costly and time-consuming to implement, active restoration in boreal forest can be implemented using standard forestry planting protocols, without any ground preparation, independently of the forest degradation state.

Key words: balsam fir, insect disturbance, moose browsing, Newfoundland, protected area, scarification

Implications for Practice

- Ecological restoration enables the protection of ecological integrity in protected areas.
- Mixed-effect models show a minimal effect of ground preparation treatment on the success of planted seedling for the restoration of balsam fir forest.
- Recommendations for balsam fir boreal forest restoration based on early seedling performance over 2 years are to plant seedlings following standard forestry planting protocols regardless of the forest degradation state.

Introduction

Global forests are being degraded at an increasing rate, with a net loss of 1.5 million km^2 over a period of 12 years (2000–2012; Hansen et al. 2013); hence ecological restoration becomes an integral component of ensuring future forest health, biodiversity conservation, and community livelihood (Lamb et al. 2005; Chazdon 2008; Lamb 2015). The importance of implementing restoration efforts with specific conservation goals to overcome failed regeneration is crucial to uphold ecological integrity targets for protected areas (Parks Canada 2008; Keenleyside et al. 2012; Wiens & Hobbs 2015). This is especially true given the challenges faced by protected areas, such as external pressures, fragmented landscape, and species diversity representation (Rodrigues et al. 2004; DeFries et al. 2005; Gaston et al. 2008; Leroux & Kerr 2013). To ensure restoration success, consideration must be given to ecological, social, and financial aspects at the planning stage (Hobbs et al. 2014). For instance, Shepperd and Fairweather (1994) found that restoration of aspen forest in Arizona requires longer periods of fencing than expected, due to heavy elk browsing and trampling, highlighted the possible need for culling. Therefore, experimentation within an adaptive management framework to select the most effective restoration protocol will best foster restoration success.

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Forests around the world are home to a high diversity and abundance of herbivores. Where herbivore densities are high, selective browsing of preferred species can create a new disturbance regime (Hobbs 2006) resulting in legacy effects that may impede natural regeneration. Such effects are especially severe on islands where ungulates were introduced (e.g. Anticosti Island, Côté et al. (2014); Haida Gwaii, Vourc'h et al. (2001); Isle Royale, McInnes et al. (1992); Newfoundland, McLaren et al. (2004); and New Zealand archipelago, Wardle et al. (2001)). Changes in environmental conditions in these degraded forests make passive recovery inadequate and active restoration a preferred solution (Yates et al. 2000; Tremblay et al. 2007; Gosse et al. 2011; Hidding et al. 2013; Côté et al. 2014). On the island of Newfoundland (Canada), moose (*Alces alces*) were introduced in the early 1900s with their number increasing quickly (150,000 moose island-wide by 1960) resulting in the severe browsing of the young regeneration layer of the balsam fir (*Abies balsamea*) forest (McLaren et al. 2004; Gosse et al. 2011). When combined with the natural forest disturbance regime (e.g. insect herbivory and wind blowdown), recruitment is severely limited due to insufficient transition to adult reproductive trees leading to the state of "spruce-moose meadows" in some areas (Charron & Hermanutz 2016). In an effort to increase balsam fir and hardwood regeneration and restore ecological integrity within their protected areas in Newfoundland, Parks Canada implemented a recreational moose hunt beginning in the fall of 2011 within their two Newfoundland national parks. The license quota for the protected areas is designed to decrease the moose density to that under natural levels of predation by wolf (*Canis lupus*), which was extirpated from the island in the early 1930s (Parks Canada 2010).

The legacy effect generated by moose browsing suggests that active ecological restoration action (e.g. artificial regeneration) in highly degraded areas should be implemented (Gosse et al. 2011; Charron & Hermanutz 2016). Very few studies have addressed the restoration of natural balsam fir forests as the focus has been on perfecting silviculture practice for industrial forestry, in particular black spruce (*Picea mariana*). Previous restoration projects in Quebec and Newfoundland have examined how to optimize survival, growth, and browsing occurrence of balsam fir seedlings (Beguin et al. 2009*a*, 2009*b*; Humber & Hermanutz 2011; Faure-Lacroix et al. 2013; Côté et al. 2014). Environmental conditions, such as the surrounding plant community, moisture, and nutrient availability, have also been documented as important factors affecting tree seedling establishment success (Brooker et al. 2006; Gómez-Aparicio et al. 2008; Prévosto & Ripert 2008; Prévosto et al. 2010; Hibsher et al. 2013; Palik et al. 2015). To improve environmental conditions, ground preparation techniques, such as scarification can be used, which affect ground structure by exposing mineral soil and mixing organic layer (Löf et al. 2012). Scarification may improve seedling survival by increasing soil moisture, temperature, and nutrient availability, reducing compaction and controlling competing vegetation (Prévost 1992; Löf et al. 2012). In Mediterranean woodlands, Aleppo pine (*Pinus halepensis*) recruitment was enhanced by the soil scarification that increased moisture and volume accessible to the root system in the ground (Prévosto & Ripert 2008). Depending on the primary role of the surrounding vegetation, protection, or competition, its suppression may have a positive or a negative effect (Brooker et al. 2006; Gómez-Aparicio et al. 2008; Wright et al. 2014). Parent et al. (2003) found that removal of the aboveground vegetation reduces aerial competition without disrupting the moist seedbed that is preferred by balsam fir; however, little else is known about the effect of ground preparation on balsam fir seedlings.

Hobbs et al. (2014) advocate for restoration planning at the landscape level, to more effectively use resources on the full spectrum of degraded ecosystems and achieve multiple management goals. In this study, we tested various experimental planting protocols in Terra Nova National Park (TNNP) with the target of reestablishing the foundation species, balsam fir. The forest landscape in TNNP is a mosaic of different disturbance intensities caused by insect disturbance, wind, and moose browsing (Charron & Hermanutz 2016). The disturbance gradient includes: (1) closed canopy mature balsam fir forest; (2) small; (3) medium; and (4) large canopy gaps created by spruce budworm (*Choristoneura fumiferana*)–hemlock looper (*Lambdina fiscellaria*) insect disturbance (Charron & Hermanutz 2016). Charron and Hermanutz (2016) found differences in plant community, light availability, soil chemistry, and physical attributes along this disturbance gradient; therefore restoration was implemented across the disturbance gradient to compare experimental planting techniques and permit subsequent adaptive management.

The aim of restoration efforts in TNNP is to eventually reestablish closed canopy balsam fir forest stands with all the inherent ecosystem functions and biodiversity. To begin the process of forest restoration, the foundation species balsam fir seedlings were experimentally planted across a disturbance gradient from closed canopy forest to large insect-disturbed stands. The objective of the study is to assess scientifically sound procedures for the reestablishment of balsam fir in TNNP forests, by evaluating the effect of (1) aboveground vegetation removal and (2) ground scarification on seedling performance (survival, growth, and browsing intensity). We hypothesize that vegetation removal and ground scarification will enhance early seedling establishment and growth, but may increase the visibility for browsing. Performance was also experimentally evaluated using low (5,000 seedlings/ha) and high (20,000 seedlings/ha) planting densities; however, density-dependent factors were not anticipated for at least 5 years (Scott et al. 1998) and since no biologically meaningful density effects were found after 2 years (Charron 2016), results are not included.

Reporting the outcomes of the first several years is crucial as this early life history time frame sets the template for the long-term success of restoration, especially long-lived forest restoration and enables us to determine the key processes that affect seedling success. The outcomes of this study will contribute to best practices to use for adaptive restoration across the disturbance gradient experienced in balsam fir forest of TNNP. This study will support broader conservation management initiatives such as lichen, mammal, and bird conservation (e.g. *Erioderma pedicellatum*, Newfoundland Marten [*Martes americana atrata*], and Newfoundland Red Crossbill

[*Loxia curvirostra percna*]); all species that depend on closed canopy forest and its resources. In addition, reinstatement of closed canopy forest will assist in minimizing the invasion of non-native plants such as Canada thistle (*Cirsium arvense*; Humber & Hermanutz (2011)).

Methods

Study Site

The experimental restoration was implemented in TNNP (48°30′N, 54°00′W), a protected area of approximately 400 km^2 located in Newfoundland, Canada. The climate is maritime with a mean temperature of −6.8 and 16.1°C in January and July, respectively, and mean annual precipitation of 311.0 cm and 872.7 mm of snow and rain, respectively (Environment Canada 1971–2000). TNNP boreal forests are dominated by black spruce inland and balsam fir along the coast. Balsam fir forest covers 15% of the park and is restricted to richer soil, predominantly humo-ferric podzols (Deichmann & Bradshaw 1984). In the mid-1990s, at the peak of their population, the density of moose exceeded carrying capacity as shown by decreasing recruitment (Mercer & McLaren 2002); the number of moose has decreased in the last 15 years leaving persistent and significant legacy effects across the park. Prior to heavy moose browsing, the balsam fir forest was naturally disturbed by insect herbivory, creating a heterogeneous matrix of regenerating to mature stands (Baskerville 1975). The intensification of browsing negatively affected regeneration and created savannah-like meadows (gaps) following insect disturbance (McLaren et al. 2004). Since the implementation of the moose hunt in 2011, an increase in hardwood regeneration and lower browsing has been observed suggesting that active restoration action is possible (J. Feltham 2016, TNNP, NL, personal communication). For a complete description of the environmental conditions at the restoration sites, see Charron and Hermanutz (2016).

Experimental Design

Twenty-four plots of 24 × 24 m were planted with balsam fir seedlings in July 2013. Plot location was randomly assigned within each site. Between 251 and 1,156 seedlings were planted per plot, a function of planting density (5,000 and 20,000 seedlings/ha) and terrain quality, for a total of 9,685 seedlings. Seedlings were 3–4 years old, ranging in height from 7.6 to 48.7 cm and were grown at Wooddale Provincial Tree Nursery (Newfoundland, Canada). To decrease moose attraction to the seedlings (Ball et al. 2000), fertilizer was only applied for the first 2 years of growth. Seeds were originally collected in Port Saunders, Newfoundland. Exceptionally, there was no precipitation in early July 2013, and dead-by-drought seedlings were replaced ($N = 596$) 2 weeks after planting. Survival and growth were assessed for seedlings and the mortality estimates did not include the first set of seedlings that died due to transplant shock.

Table 1. Experimental restoration planting protocols established in Terra Nova National Park (Newfoundland, Canada) in 2013. Controls were established in closed canopy (BHCC, Blue Hill Closed Canopy), small (BCB, Bread Cove Brook), medium (PLC, Platter's Cove), and large insect gaps (BH, Blue Hill), while treatments (ground and density) were only tested in Blue Hill Closed Canopy and large insect gap at Blue Hill. For medium and large insect gaps, the edge and center plots were pooled, resulting in four replicates.

Site	*Ground Treatment*	*Density (Seedlings/ha)*	*Number of Plot Replicates*	*Range of Planted Seedlings per Plot*
BHCC	Control	5,000	2	256
		20,000	2	1,156
	Aboveground cut	5,000	2	256
	Scarification	5,000	2	256
BCB	Control	5,000	2	252–256
PLC	Control	5,000	4	256
BH	Control	5,000	4	251–256
		20,000	2	1,110–1,156
	Aboveground cut	5,000	2	255–256
	Scarification	5,000	2	256

Seedlings were planted at four different sites along a gradient of disturbance encompassing: (1) closed canopy mature balsam fir forest; (2) small (<5 ha); (3) medium (approximately 15 ha); and (4) large (approximately 50 ha) canopy gaps created by insect disturbance; no moose use difference was observed along the gradient (Charron & Hermanutz 2016). Within medium and large canopy gaps sites, plots were located at the center and the edge of the opening, but mixed-model analysis found no difference in environmental conditions and seedling success between center and edge; therefore, location (center/edge) were pooled and not considered as a variable in the analyses (Table S1, Supporting Information).

The ground and density treatments were randomly established at the extremes of the disturbance gradient (closed canopy and large opening) because of limited time and resources. We used closed canopy forest as a control as environmental conditions were similar to undisturbed balsam fir forest (Charron & Hermanutz 2016). Prior to seedling planting (June 2013), three ground treatments were applied: (1) control (no ground treatment); (2) aboveground treatment (all aboveground vegetation over 10 cm high was cut for the entire plot); and (3) soil scarification (combined with aboveground plant removal prior to scarification). At the location of each seedling, the soil was scarified with a Pulaski ax approximately 15 cm belowground and approximately 30 cm radius around the seedling. All combinations of site/treatment/density were replicated at least twice (Table 1). It was not a fully factorial design, with no interaction between ground treatment and densities (Table 1). At the small and medium gaps sites, all seedlings were planted using the control ground treatment (i.e. no intervention) at 5,000 seedlings/ha. The use of control ground treatment at the four sites acts as a baseline response across the disturbance gradient.

Plant Community Structure. In TNNP, the plant community is highly correlated with environmental conditions, and is therefore a good proxy of environmental conditions (Charron & Hermanutz 2016). For each restoration plot, percent cover of all species less than 2 m was estimated to the nearest 5% in 5×1-m^2 quadrats located equally along a diagonal transect running between two opposite corners (4 sites, 24 plots $\times$ 5 quadrats; $n = 120$ plots) in June 2013. Grass species were grouped to family level (Poaceae), and rare or unidentifiable species at time of survey (e.g. species that were not flowering) were identified to genus. The presence of palatable species (e.g. *Viburnum nudum*, *Betula papyrifera*, *Cornus stolonifera*, and *Taxus canadensis*) within 50 cm of the seedling was recorded, following Pimlott (1953) and Tanner and Leroux (2015).

Evaluating Seedling Success. Individual seedlings were surveyed after the first growing season (October 2013), after winter (May 2014), and after the second growing season (October 2014). Seedling survival, growth, and browsing occurrence were recorded at each survey date. Survival was recorded as a binomial variable (alive or dead). If the seedling was dead, the cause of death was recorded; drought, herbivore (moose, hare [*Lepus americanus*]), or unknown. Seedling growth traits recorded were total height (cm), growth length of the terminal leader (cm), basal diameter (mm), and number of buds on the terminal leader. Total height was measured from the ground to the tip of the leader. New growth was measured from the last year bud scar to the tip of the leading branch, excluding buds. Basal diameter was measured with a caliper at the root collar. In addition, the initial number of branches and, for more complex tree structure, the initial number of possible leader was recorded. Browsing was recorded as a binomial variable, either present or absent, with no differentiation between old and new browse events. The identity of the browser (moose or hare) was also recorded.

Statistical Analysis

Plant Community Structure. For modeling purposes, each species percent cover was summarized using ordination technique on the understory plant survey data, and ordination scores used as proxy of environmental conditions. Principal component analysis (PCA) was performed using the "rda" function of the Vegan package (Oksanen et al. 2015). To ensure adequate data redundancy (Peck 2010), mosses were pooled by genus for analysis, which does not affect results as the dominant mosses were monospecific. Due to nonlinearity and a zero-rich dataset, the same dataset was analyzed with the "metaMDS" (non-metric multidimensional scaling [NMDS] analysis) and a procrustean analysis was performed to ensure consistency. The procrustean superimposition approach overlays two ordinations solutions, rotates, and scales them to optimize their fit (Jackson 1995). In addition, a permutation procedure was done to evaluate the significance of the fit (Jackson 1995; Peres-Neto & Jackson 2001). The procrustean superimposition analysis was performed with the function "procrustes" and significance was tested with the "protest" function in R.

Seedling Success Modeling. To test the effect of the various restoration treatments on seedling success, we constructed mixed-models with plot as a random effect ($n = 2$ or 4; Table 1), accounting for spatial autocorrelation. We used logistic regression with seedling survival and browsing occurrence on the living seedlings as our response variables and linear model with new growth length, basal diameter, and number of buds as our response variables to predict seedling success according to uncorrelated covariates: PCA scores, presence of palatable species in the area surrounding the seedling, initial seedling height, and number of leaders, planting density and ground treatment (Table S2). For each model, correlation coefficients between explanatory variables were $|r|$ less than 0.404 ($n = 24$ plots, $p < 0.05$; Upton & Cook (2008)) signifying there were no significant correlations between any treatment and the plant community (PCA scores and palatable species presence). To extract the effect of each treatment, models were sequentially constructed by ordering the covariates, with the restoration treatment as the last covariate ((1) plant community; (2) individual seedling traits; and finally, (3) treatment). All plots were used, even with the absence of ground treatment and varying planting density in the small and medium insect opening (Table 1) as inclusion allowed the best resolution of seedling success under the conditions of field planting along the entire disturbance gradient. Models were constructed using the functions "glmer" (logistic regression) and "lmer" (linear mixed-models) of the "lme4" packages (Bates et al. 2015). Confidence intervals (95% CI) were computed for the treatments coefficient estimates, and significance level was set such that the CI does not overlap 0.

To evaluate the importance of plant community and individual traits on seedling success, we performed model selection on a constant set of candidate models (Table S3) and determined the models with the most of the evidence based on Akaike's information criterion (AIC). Marginal and conditional goodness-of-fit (R^2) was computed following the procedure of Nakagawa et al. (2013). Models with pretending variables were eliminated from the AIC selection routine. Summary of the best predictive models allowed us to determine the important covariate(s) affecting seedling success.

Results

Plant Community Structure

A high percentage of the variance in the plant community was explained by the first two axes (53.3 and 21.1% variance explained, respectively) of the PCA. The procrustean superimposition of the PCA and NMDS ordination techniques gave a high fit ($m_{12} = 0.2517$, $p = 0.001$); therefore, PCA scores derived from plant cover data can be used with confidence as a proxy for environmental conditions. The first axis represents a gradient from early succession communities, adapted to open, warm, and dry conditions, to shade tolerant and moist adapted communities, while the second axis shows a gradient from communities with deciduous species, to coniferous and feathermoss dominated communities. PCA results and all axis scores for

Table 2. Summary of balsam fir seedling success (mean ± SD) after two growing seasons along a gradient of disturbance (closed canopy, small, medium, and large insect gaps) in Terra Nova National Park (Newfoundland, Canada). Proportion of browsed seedling was calculated for all seedlings (dead or alive) after the two growing seasons.

	Overall	*Closed*	*Small Opening*	*Medium Opening*	*Large Opening*
Number surveyed	7,316	2,692	461	1,024	3,139
Proportion dead (%)	9.5	3.6	11.9	10.7	13.8
Drought	6.1	2.4	7.8	4.8	9.4
Moose	3.2	1.2	3.4	5.6	4.2
Other (hare, unknown)	0.2	>0	0.7	0.3	0.2
Proportion browsed (%)	8.1	2.0	14.3	15.7	9.8
New growth increment (cm)					
2013	4.8 ± 2.5	4.7 ± 2.4	4.7 ± 2.4	4.9 ± 2.7	4.9 ± 2.5
2014	1.2 ± 1.5	0.6 ± 1.0	0.8 ± 1.0	2.1 ± 1.8	1.5 ± 1.6
Basal diameter increment (mm)					
2014	0.63 ± 0.73	0.24 ± 0.61	0.61 ± 0.70	1.13 ± 0.65	0.85 ± 0.68
Number of leader buds					
2013	2.9 ± 1.0	2.9 ± 1.0	3.0 ± 1.0	3.1 ± 0.9	2.9 ± 1.0
2014	2.0 ± 1.0	1.4 ± 0.7	1.7 ± 0.8	2.6 ± 0.8	2.5 ± 0.9

the two axes are included in Supporting Information (Table S4; Fig. S1).

Seedling Success

Seedling survival was very high after the first two growing seasons (90.5%; Table 2), with less than 2% (75) of the seedlings not relocated during the survey. Cause of death was mainly attributed to drought (6.1%), with small loss to moose browsing (3.2%) and other causes (0.2%; Table 2). Growth, measured as number of buds, new growth, and basal diameter increment, was dependent on the year, with better growth in 2013 than 2014 (Table 2). Browsing appears to be minimal; over the two growing seasons, 8.1% had been browsed by October 2014, all by moose, except one occurrence of hare browsing and one unknown cause. Closed canopy stands had better survival and fewer browsing occurrences, but lower growth compared to the insect-disturbed areas (Table 2).

Effect of Treatment on Seedling Success

The impact of the restoration treatments was minimal, with only the scarification treatment showing statistically significant effects, compared to the control conditions of field planting. Contrary to prediction, the sequentially constructed mixed-model shows that soil scarification diminished the seedling's survival after the first growing season (95% CI: −1.9, −0.5; Fig. 1). However, the effect of scarification is dependent on the environmental conditions, and a marginally better survival than the control treatment is seen in the closed canopy site that has preserved the environmental conditions of a mature balsam fir forest (scarification: 98% vs. control: 95%, by October 2014; Fig. 1). In the open deforested area, a negative effect on survival is experienced by seedlings (scarification: 82% vs. control: 86%, by October 2014; Fig. 1). After the second growing season, scarification treatment does not affect survival, but differences from the first growing season are still present (Fig. 1). Growth measures (length of the new terminal growth,

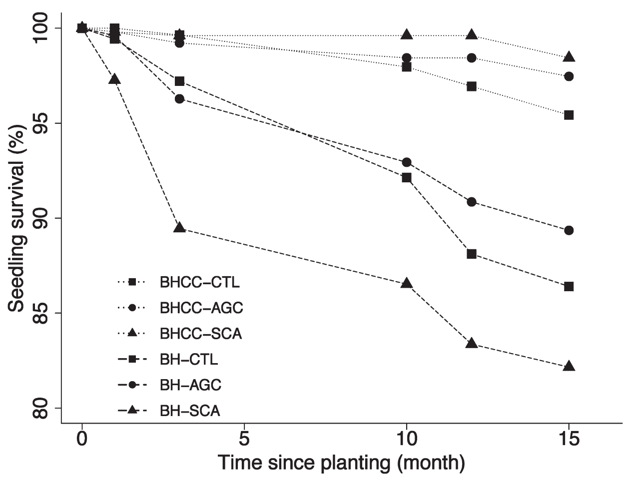

Figure 1. Survival curves of balsam fir seedlings after two growing seasons in Terra Nova National Park (Newfoundland, Canada). Each site is identified by line type (dashed: BH, large insect gap; dotted: BHCC, closed canopy mature forest) and each ground treatment by a symbol (square: CTL, control; circle: AGC, aboveground cut; triangle: SCA, scarification). The average survival curve equation (logarithmic) for BH is $y = -4.32\ln(x) + 100$ (R^2: 0.87) and for BHCC is $y = -0.911\ln(x) + 100$ (R^2: 0.70).

number of buds on the leader and basal diameter) increased only marginally with the scarification treatment after the second growing season (control: 1.1 cm, 2.0 buds, 0.6 mm; scarification: 1.7 cm, 2.2 buds, 0.8 mm, respectively; Fig. 2). Contrary to predictions, only minimal effects of the aboveground treatment were seen on overall seedling success. As well, browsing was not affected by any experimental restoration treatment.

Selection of Predictive Models for Seedling Success

Because the effect of the active restoration treatments tested was minimal, individual seedling traits and the plant community (PCA score and presence of palatable species) were examined

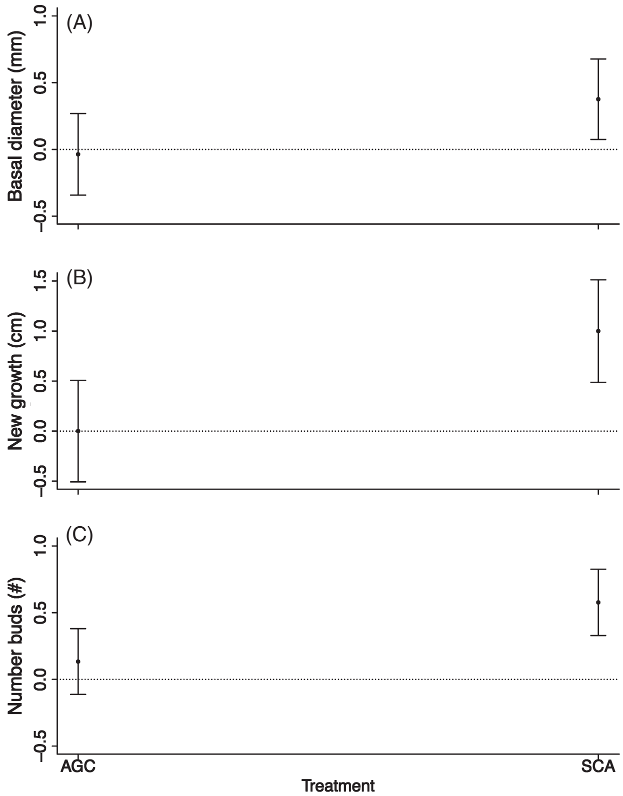

Figure 2. Coefficient confidence intervals (95% CI) for the ground treatment compared to control planting represented by "0" dotted line (AGC, aboveground removal; SCA, scarification) on (A) basal diameter, (B) new growth, and (C) number of buds after the second growing season in Terra Nova National Park (Newfoundland, Canada). Coefficients come from sequentially constructed models: Y = vegetation + individual + ground. Significant effect when the ±95% CI does not overlap 0 (control—dotted line).

to predict seedling success using AIC selection. Top models for each seedling success attributes (growth, survival, and browsing) and monitoring time are summarized in Table 3. The relative importance of individual seedling traits and plant community varied as a function of the response variable and time since planting (Table 3; Fig. 3). Individual seedling traits explained a small proportion of the growth pattern after the first growing season (new growth R^2: 7.1%, number of buds R^2: 16. 2), with taller seedlings at planting having a greater new growth and more buds. However, smaller seedlings with multiple leaders had a better survival following the first growing season than taller seedlings with only one leader.

Plant community (PCA score and palatable species presence) was driving the second growing season growth (new growth R^2: 11.8%, number of buds R^2: 24.3, and basal diameter R^2: 19.7%). Seedlings planted under closed canopies and in moist environments and close to palatable species had a better survival than those planted in open and dry areas, but the latter seedlings grew marginally better, with greater new growth, more buds, and greater basal diameter (Table 2). For the seedlings that were still alive after the second growing season, the chance of being browsed was explained by the environmental conditions and the individual traits (initial height) of the seedling (R^2: 29.8 and 11.1%, respectively), with taller seedling growing in open and dry environments having a higher incidence of browsing. Therefore, better growth, lower survival, and higher browsing intensity were observed with increasing forest disturbance.

Discussion

Ecological restoration is becoming a key part of forest conservation; however, to be effective, restoration must be based on scientifically sound protocols. Our study showed, that although statistically significant, no biologically significant impacts of ground interventions on seedling success; after two growing seasons there was no increase in survival or growth, or a reduction in browsing pressure. Contrary to our hypothesis, ground treatments did not have a consistent positive effect on seedling success, especially in focal restoration areas that had undergone significant degradation, the large gaps.

Considering the cost in both financial and human resources needed to implement ground restoration treatments, it seems most cost-efficient to limit invasive treatments such as scarification, and simply plant seedlings into the surrounding communities using the standard forestry tree planting protocols. As well, for restoration efforts within protected areas, limiting the human impact created by ground treatment is advantageous and may limit non-native plant invasions (Hobbs & Huenneke 1992). Despite the interaction between the environment and ground treatment; with (1) improved growth, but lower survival in degraded areas and (2) improved growth and survival in closed canopy forest, amelioration provided by the ground treatments were very small; hence the recommendation of using the control treatment of field planting. Therefore, based on the high survival and growth results of the first 2 years, restoration of balsam fir forest can be implemented using field planting in the park, independently of the past disturbance regime.

Despite the observed ground physical improvement and successful revegetation observed in previous restoration studies, which have been mostly carried out in dry Mediterranean sites and for broadleaf species (Löf et al. 2012), ground treatment did not have the expected positive effect on the success of our experimentally planted seedlings. Moreover, soil scarification resulted in lower survival in open and degraded areas; the highest priority stands for the ecological restoration in TNNP (Charron & Hermanutz 2016). Results are consistent with Beguin et al. (2009*b*), where mechanical scarification leads to disturbed understory vegetation, soil desiccation, and modified microclimate in a boreal forest. In the closed canopy forest, scarification did not have the negative effect seen in open and disturbed area because of the moister conditions provided by the shaded forest and the thick feathermoss seedbed (Place 1956; Côté & Bélanger 1991; McLaren & Janke 1996; Parent et al. 2003). Aboveground removal treatment had no significant effect, compared to the control, showing that aboveground competition for light is not a limiting factor for the shade-tolerant balsam fir seedlings (Messier et al. 1999; Duchesneau et al. 2001). Moreover, it was noted that the aboveground removal

Table 3. Model selection results on a constant set of candidate models (full list of models, Table S3) for all seedling success variables. Vegetation (Veg) refers to the first and second PCA axis score and the presence of palatable species; individual traits (Ind) refer to height and number of leader at planting; ground (Gr) refers to ground treatment; and density (Den) refers to planting density.

Variable	*Date*	*Model*	K	*AICc*	*Delta AICc*	*AICc Weight*	*Log Likelihood*	*Marginal* R^2	*Conditional* R^2
Survival	October 2013	Veg + Gr	7	1,469.11	0.00	0.89	−727.55	0.23	0.28
		Veg	5	1,473.36	4.25	0.11	−731.68	0.19	0.29
	May 2014	Veg + Ind	8	2,960.08	0.00	1.00	−1,472.03	0.23	0.27
	October 2014	Veg + Ind	8	4,186.05	0.00	1.00	−2,085.01	0.16	0.21
Browsing	October 2014	Veg + Ind	8	2,293.74	0.00	1.00	−1,138.86	0.36	0.45
New growth	May 2014	Ind + Gr	8	31,448.88	0.00	0.48	−15,716.43	0.07	0.09
		Veg + Ind	9	31,449.47	0.59	0.36	−15,715.72	0.07	0.09
		Ind	6	31,451.00	2.12	0.17	−15,719.49	0.07	0.09
	October 2014	Veg + Ind + Gr	11	22,367.07	0.00	0.98	−11,172.52	0.18	0.26
Basal diameter	October 2014	Veg + Den	7	804.59	0.00	0.52	−395.16	0.23	0.33
		Veg + Gr	8	805.71	1.12	0.29	−394.68	0.23	0.33
		Veg	6	806.58	1.99	0.19	−397.19	0.20	0.32
Buds	October 2013	Veg + Ind	9	18,710.20	0.00	0.87	−9,346.09	0.16	0.17
		Ind	6	18,713.95	3.75	0.13	−9,350.97	0.16	0.17
	October 2014	Veg + Ind + Gr	11	15,703.95	0.00	1.00	−7,840.95	0.29	0.34

treatment had a shorter-lasting effect than the scarification treatment on the suppression of the competing vegetation (Prévosto & Ripert 2008), which was dominated by herbaceous species in the open areas. Ground treatments did not affect the browsing intensity, despite the visibility given to seedlings. Moose preference for taller saplings, seedling protection provided by the snow cover and/or effective moose reduction from the recreational hunting program could explain the low browsing occurrence (8.1%, browsing of <60 cm seedlings [this study] compared to 97% > 60 cm (Gosse et al. 2011)) and limited effect of vegetation protection (Brooker et al. 2006; Gómez-Aparicio et al. 2008).

The main factors affecting seedlings success were related to seedling traits and the surrounding plant community. As demonstrated by Faure-Lacroix et al. (2013), taller seedling stock had higher growth, but also experienced increased browsing pressure. Restoration work could plant taller seedlings to increase success; but planting older, larger seedlings could result in higher browsing as well as being a more expensive intervention, and is a trade-off that is project dependent and would need to be evaluated for each situation (Newton et al. 1993; South & Mitchell 1999). As well, the conditions in which seedlings are planted in the landscape affect their success. However, we found that growth of our fir seedlings after the first growing season was excellent, and differences may be masked by the good conditions experienced by the seedlings in the prior year spent in the nursery; hence continued monitoring is needed to evaluate this effect. Along the disturbance gradient, from closed canopy forest to large open areas, conditions spanned from moist and shaded, to warm, dry and open, affecting survival, growth and browsing occurrence of the seedlings. The highest priority stands for restoration are the large and medium open areas (Charron & Hermanutz 2016), which experienced lower survival, higher growth, and more browsing pressure than the other areas. Variable seedling responses, with more or less growth and survival, are observed across the landscape; hence varying the minimal planting density can be used in a cost-effective manner to reach the optimal forest stand stocking. The target adult density needed to regenerate a closed canopy forest, based on similar boreal forests would be approximately 2,500 trees/ha (Greene et al. 1999; Tremblay et al. 2007). Based on the average survival curve (logarithmic function) of this study, for both areas, the density to reach 2,500 trees/ha after 20 years would need to be 2,570 seedlings/ha in the closed canopy and 2,870 seedlings/ha in the disturbed large area, which is lower than the planting density used in this study. However, drought conditions would need to be incorporated into the determination of planting density.

The statistical models constructed to predict seedling success in function of restoration treatments, plant communities, and individual traits have a high proportion of unexplained variation, underscoring the need for further research. Sources of variations could be due to: (1) the genetic variation, only assessed indirectly through individual seedlings traits (Thomas et al. 2014) and (2) the microhabitat heterogeneity, not represented by the large size of the plots (Ameztegui & Coll 2015). Ameztegui and Coll (2015) found that seedling browsing intensity and mortality was partially explained by microhabitat factor, represented by the distance to shrubs. In this study, the high seedling survival, and absence of difference between aboveground vegetation suppression treatment and control plots, as well as a general scarcity of shrubs throughout TNNP landscape, suggests that facilitation by shrubs is not an important factor. In terms of planting density for restoration, early density response was not anticipated; however, further surveys and comparison with a broader range of densities will establish if density-dependent effect are seen on competition and browsing intensity, affecting seedlings success.

In conclusion, as moose negatively affect large tracts of forested areas, implementing effective active restoration strategies are crucial in reaching restoration targets in Newfoundland. The experimental restoration tested in this study is part of an adaptive management framework and indicates that often-used planting manipulations, such as intensive ground treatments, are not needed in all restoration efforts and should be evaluated prior to initiating large-scale projects. Planting seedlings following tested standard forestry planting

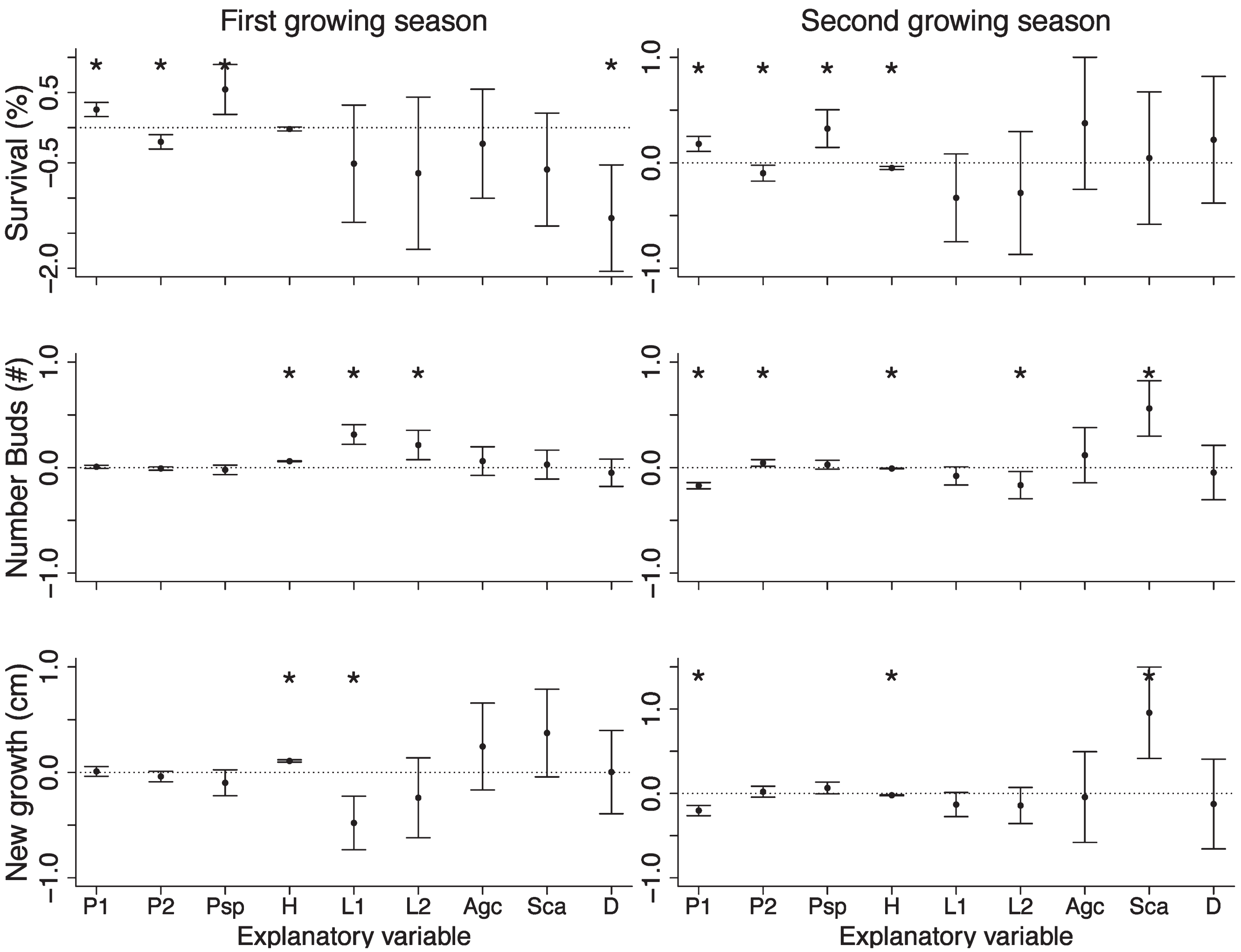

Figure 3. Coefficient confidence intervals (95% CI) for all variables under studies: P1/2, PCA axis 1/2; Psp, presence of palatable species; H, height at planting; L1/2, presence of 1/2 leader(s) compared to >2 leaders; Agc, aboveground cut treatment; Sca, scarification; D, high density in Terra Nova National Park (Newfoundland, Canada). Coefficients come from sequentially constructed models: Y = vegetation + individual + ground + density. Significant effects when the ±95% CI does not overlap 0 (dotted line) are indicated by a star (*).

protocols is the recommended practice for balsam fir forests in Newfoundland.

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Supporting Information

The following information may be found in the online version of this article:

Table S1. Mixed-model results of balsam fir seedling success as a function of the planting location (center, edge) in Terra Nova National Park (Newfoundland, Canada).
Table S2. Description of explanatory variables used for the construction of mixed-models following AIC model selection procedures.
Table S3. Candidate models for AIC selection of the best models explaining balsam fir seedling success planted in Terra Nova National Park (Newfoundland, Canada).
Table S4. Scores of the two first axes of the principal component analysis (PCA) on the understory plant community of study sites in Terra Nova National Park (Newfoundland, Canada).
Figure S1. Ordination scatterplot (principal component analysis; PCA) of plant community in Terra Nova National Park (Newfoundland, Canada).

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